Ref. 10

Origin in Tobacco Smoke of N′-Nitrosonornicotine, a Tobacco-Specific Carcinogen: *Brief Communication* [1,2,3]

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ABSTRACT—To evaluate risk factors and to approach methods of reduction of the carcinogenic potential of cigarette smoke, the transfer rate of N-nitrosonornicotine in a popular U.S. blended cigarette into mainstream smoke was quantitatively determined. The mean transfer rate was 11.3%; thus ≈46% of the tobacco-specific carcinogen in the smoke came from the tobacco, and the remainder was synthesized during smoking.—J Natl Cancer Inst 58: 1841–1844, 1977.

Tobacco chewing and smoking are associated with cancer of the oral cavity and esophagus (*1–5*). NNN has been identified in tobacco products and shown to be an esophageal carcinogen in rats (*6*). Concentrations range from 0.3 to 90 ppm in tobacco (*7–10*) and from 140 to 250 ng in the smoke of one cigarette (*11*). These levels are high compared to nitrosamine concentrations in meat and other food products, which rarely exceed 0.1 ppm (*12*). NNN was not detected in freshly harvested tobacco (<1 ppb) but was formed during tobacco processing (*13*). The presence of NNN in smoke can be explained either by its direct transfer from tobacco into smoke and/or by its formation during smoking. If a significant portion of NNN in cigarette smoke originated by transfer from tobacco, appropriate steps could then be taken to inhibit its formation during tobacco curing and processing.

MATERIALS AND METHODS

^{14}C-Labeled NNN (*14*) was used as a radioactive tracer for NNN transfer into mainstream smoke, and [^3H]NNN was used as the internal standard. The analytic procedure is a modified version from an earlier published method (*15*).

Apparatus.—A Packard 3385 Tri-Carb was used for the scintillation counting. Discriminator settings were optimized for the most efficient simultaneous counting of ^{14}C and ^3H labels in a scintillation cocktail of 0.4% of PPO (2,5-diphenyloxazole) and 0.005% POPOP (1,4-bis[2-(5-phenyloxazolyl)]benzene) in toluene. For analysis, we used a model ALC/GPC-202 high-speed liquid chromatograph (Waters Associates, Inc., Milford, Mass.) with a 254-nm differential ultraviolet detector, a Hewlett-Packard gas liquid chromatograph 5710 with a flame ionization detector, and a Hewlett-Packard model 5980A mass spectrometer. The cigarettes were smoked with a Borgwaldt 20-port smoking machine, on which every second smoking channel of the rotating head was connected to a nitrogen source (flow rate, 17.5 ml/sec) (*7, 11*).

Reagents.—All solvents were spectrograde. [2′-^{14}C]NNN (4.1 mCi/mmole) and unlabeled NNN were synthesized as described previously (*14*). Basic alumina (activity II–III ICN-Pharmaceuticals, Inc., Cleveland, Ohio) was used for chromatography. Carbowax 20-M TPA (10%) on Chromosorb W (mesh, 80–100) was obtained from Applied Science Laboratories, Inc. (State College, Pa.). [^3H]Nornicotine (45 mg, 0.30 mmole, 296 mCi), obtained by catalytic tritium exchange (New England Nuclear Corp., Boston, Mass.) of nornicotine, was dissolved in 3.0 ml of 1 N HCl. After dropwise addition of 1 ml solution of $NaNO_2$ (700 mg, 10.1 mmole), the resulting mixture was allowed to stand at room temperature for 24 hours; it was then brought to pH 12 with aqueous NaOH and extracted three times with 10 ml of $CHCl_3$, dried (Na_2SO_4), and concentrated. It yielded a residue which was purified by preparative thin-layer chromatography on silica gel with elution by chloroform–methanol (9:1) (R_f of NNN = 0.60). This [^3H]-NNN (42% yield; >99% purity according to GLC analysis) had a specific activity of 260 mCi/mmole. In order to obtain material of constant specific activity with respect to the analytic procedure, a mixture of [^3H]NNN and nicotine was carried through the entire analytic process as described below, and the [^3H]NNN was isolated in the final step by HPLC. This procedure was repeated to give [^3H]NNN a constant specific activity (210 mCi/mmole), which was then used as internal standard in the transfer rate study of [^{14}C]NNN.

Cigarettes.—The nonfilter cigarettes (85-mm) were bought on the open market in Westchester County, New York in the winter of 1975–76. Two hundred cigarettes were opened, and 20-g samples of their tobacco (moisture content, 11.0%) were used for tobacco analysis. For the smoke analysis of cigarettes treated with [^{14}C]NNN 50 weight-selected cigarettes (±20 mg of average weight of 200 cigarettes) were used in one experiment; in all other experiments, 100 non-weight-selected cigarettes were smoked. The standard smoking conditions were 1 puff/minute of 35-ml volume and 2 seconds' duration, to a butt length of 23 mm (*16, 17*).

Cigarettes labeled with [^{14}C]NNN.—A solution of [^{14}C]NNN in water (0.017 mg/ml) was prepared for injection with an automatic microsyringe applicator using a 50-μl Hamilton syringe. The microswitch settings of the instrument were such that 38 μl (0.66 μg [^{14}C]NNN; 3.4×10^4 dpm) of the solution was deposited in the mid-

ABBREVIATIONS USED: NNN = N′-nitrosonornicotine; GLC = gas liquid chromatography; HPLC = high-pressure liquid chromatography.

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